NovaChron

KPI, metrics and formula specification

Back-end and data engineering reference for the associate, supervisor and leadership dashboards

Item	Detail
Product	NovaChron, transaction intelligence for healthcare revenue-cycle operations
Audience	Back-end developers, data engineers, QA and business analysts
Companion file	NovaChron_Product_UI.html, the front-end prototype. KPI IDs in this document appear on the screens when Dev overlay is on.
Reference code	Functions `K()` and `G()` in the prototype implement every formula in sections 6 to 8 and can be used as a test oracle.
Version	1.0 draft for design and engineering review
Date	20 October 2026
Data status	All names, volumes, benchmarks and targets in the prototype are illustrative. Targets are configuration, not constants.

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1 Purpose and how to use this document

This document defines every number shown in NovaChron: how it is derived from raw events, the exact formula, the unit, how it rolls up from associate to team to organisation, and the rule that governs it. It is written so that a back-end team can build the metrics layer without asking the design team what a tile means.

How the pieces fit

- **One definition, many screens.** Each KPI has an ID such as `O02`. The same ID appears in the API, on the dashboard widget (Dev overlay) and in this document.
- **Time first.** Almost every ratio is built from the seven-state time model in section 4. Get that model right and the rest is arithmetic.
- **Configuration, not constants.** Standards, targets, weights and thresholds in section 5 are stored per client and process. The prototype shows the defaults used in the sample data.
- **Numerators and denominators travel.** The API returns the underlying sums so that every ratio can be recomputed after aggregation. Percentages are never averaged.

Conventions

Term	Meaning
Minute values	Stored as seconds, rolled up to minutes with largest-remainder rounding so that state totals equal logged time exactly.
Period	`today` runs from shift start to the last completed minute. `week` and `month` use calendar working days in the operating time zone.
Associate-day	One associate with at least one minute logged on one working day. Used as the unit for per-associate averages.
Standard	The contractual or engineered value for a process, such as standard handle time per transaction.
Transaction	One claim worked to a closing disposition (`claim_closed`). Partial work on a claim that is not closed in the period is counted in time, not in output.

2 System context and data flow

NovaChron observes work through a lightweight endpoint agent and workflow adapters, attributes time to claims, classifies each moment into a time state, and serves role-scoped metrics to three dashboards.

Layer	Responsibility	Key design rule
Collection	Endpoint agent and workflow adapters emit events: foreground application, input inactivity, call state, claim lifecycle, SOP checkpoints.	No screen content or keystrokes. Window class and title patterns are matched on the device against an allow-list. Claim identifiers are tokens.
Attribution	Claim context engine assigns each interval to a claim using adapter events first, then title patterns, then inference.	Every attribution carries a confidence score. Low-confidence intervals are flagged for the associate to confirm.
Time-state model	Assigns each second to exactly one of seven states using the precedence rules in section 4.	States are exclusive and exhaustive. Totals reconcile to logged time.
Metrics layer	Computes KPIs T01 to F02 per associate-day and stores numerators and denominators.	Formulas defined once, versioned, shared by all APIs.
Serving	REST for queries, server-sent events for the live floor and nudges.	Role and team scoping at the gateway. Groups under five are suppressed.

Layer	Responsibility	Key design rule
Experience	Three personas plus handoff pages.	Never recomputes a KPI in the browser.

3 Events and entities

3.1 Event envelope

Every event carries the same envelope. Payload fields depend on the type.

Field	Type	Description
`event_id`	ULID	Unique, sortable identifier. Used for idempotent ingestion.
`tenant`	string	Client organisation and site.
`device`	token	Endpoint identifier.
`associate`	token	Pseudonymous associate key. The employee ID is held only in the identity service.
`ts_utc`	timestamp	Millisecond precision, UTC. Device clock is corrected against server time at session start.
`type`	enum	See the event table below.
`app, screen_class`	string	Application name and a class from the allow-list, such as `claim_detail`. Free text is never stored.
`claim`	token	Vault token for the claim, never the claim number.
`signal, confidence`	enum, 0 to 1	How the claim was identified: adapter, title_regex or inference, and the confidence.

3.2 Event types

Event type	Raised when	Used for
`session_start, session_end`	Login and logout to the work environment	T01 logged-in time
`break_start, break_end`	Scheduled or unscheduled break recorded in the WFM schedule or tagged by the associate	T02 break time
`foreground_change`	Active window or screen class changes	Classification into value-add, business value-add or non value-add
`idle_start, idle_end`	No keyboard or mouse input for the configured threshold, and when input resumes	T07 user idle time
`app_health`	Application unresponsive, portal timeout, no work allocated	T08 system idle time
`call_state`	Call connect, hold, resume, disconnect from telephony	T06 wait and hold, explains input gaps during calls
`claim_opened, claim_closed, claim_reopened`	Claim lifecycle in the client workflow	O01 output, Q03 rework, AHT
`sop_checkpoint`	A checkpoint in the process pack was evaluated, with pass or fail	Q02 SOP adherence, nudges

Event type	Raised when	Used for
`audit_result`	Quality audit scored a claim	Q01 quality score
`tag`	Associate tags a gap, for example "waiting on payer"	Reclassifies user idle to wait or system idle after review

3.3 Core entities

Entity	Key attributes
associate	associate_key, team_id, role, ramp_profile, shift calendar
team	team_id, supervisor, process, client, associate count (suppression uses this)
process_pack	Per process: standard handle time, standard time mix (section 5), screen_class to state mapping, NVA cause codes, SOP checkpoint list, SLA rules, target overrides
time_segment	associate_key, start_ts, end_ts, state, claim token, cause code, app, confidence
claim_fact	claim token, associate_key, opened_ts, closed_ts, reopened flag, SLA due, audit score
kpi_daily	associate_key, date, KPI numerators and denominators for each KPI, process pack version
nudge	nudge_id, rule, associate_key or team, evidence JSON, message, severity, status, timestamps
points_ledger	associate_key, date, rule_id, evidence, points, status (pending, final, held)
badge_award	associate_key, badge_id, awarded_ts, evidence
fmea_register	failure mode, step, effect, cause, control, detection KPI, severity, detection, owner
recommendation, action	See section 13

4 Time-state model

Each logged second belongs to exactly one state. The seven states below are exhaustive, so logged time always equals the sum of the states.

State	Token	Definition
Value-add	`va`	Foreground work that the process pack classes as value-add for the claim, such as working the claim in the client application.
Business value-add	`bva`	Necessary work that does not itself advance the claim, such as portal status checks and compliance steps.
Non value-add	`nva`	Work that does not advance the claim: tracker updates, duplicate entry, application switching, searching, other manual work. Each NVA segment carries a cause code.
Wait / hold	`wait`	External wait with no concurrent touch work, such as a payer hold when no other claim is receiving input. Hold time on one claim while another is worked is credited to the claim being worked.
User idle	`uidle`	No keyboard or mouse input for 120 seconds or more, outside calls, breaks and system waits. Counted from the last input.
System idle	`sidle`	Application not responding, no work allocated, or portal timeout for 60 seconds or more. Never attributed to the associate.
Break	`brk`	Scheduled or unscheduled break.

4.1 Precedence rules

Evaluate in this order for each second. The first rule that matches assigns the state.

1. Outside a session: not logged. Excluded from all KPIs.
2. A break is open: `brk`.
3. A system condition has lasted at least `sIdleSec` (60 s): `sidle` for the whole interval, starting when the condition began.
4. A claim is in external wait (payer hold, portal processing, client application wait) and no other claim received input in the interval: `wait`. Sub-cause is stored for the wait pareto.
5. No input for at least `uIdleSec` (120 s), not explained by rules 3 or 4 or an active call: `uidle`, counted from the last input.
6. Otherwise foreground work: map `screen_class`, application and claim context through the process pack to `va`, `bva` or `nva`.

4.2 Derived time quantities

```
available   = logged − break                          (T03)
active      = va + bva + nva                           (T04)
productive  = va + bva                                  (T05)
handle      = active + wait                            (used for AHT)
idle        = uidle + sidle                            (T09)
invariant   : logged = brk + va + bva + nva + wait + uidle + sidle
```

> **Reconciliation check.** Run the invariant nightly for every associate-day. Any difference other than zero after rounding is a defect in the time-state model, not a data-quality note.

4.3 Dispute and review

An associate can dispute an idle episode in the associate view. A disputed episode is held out of idle and rewards calculations until the supervisor accepts it (episode stays idle) or reclassifies it (for example to `wait`). Review decisions are stored as `tag` events with the reviewer and time, so the original classification can always be reconstructed.

5 Configuration parameters

All values below are defaults used in the prototype. Store them per client and process with an effective date, and record the parameter version on every `kpi_daily` row so that history can be reproduced.

Parameter	Default	Unit	Used by
`stdAht`	9.0	min	Standard handle time per transaction. O02, O03, O05, O08.
`stdMix`	va 5.1 bva 1.6 nva 0.9 wait 1.4	min per txn	Standard minutes by state per transaction. Must sum to `stdAht`. RCA variance (section 11).
`occT`	0.85	ratio	Target occupancy. O03 expected output. Healthy band is 82 to 90 percent.
`utilT`	0.80	ratio	Target utilisation. C01 utilisation component.
`nvaT`	0.08	ratio of T03	NVA ceiling. Capacity-gap attribution, V05, badge and nudge rules.
`idleT`	0.08	ratio of T03	Total idle ceiling. V05.

Parameter	Default	Unit	Used by
`qualT, sopT`	95, 95	percent	Quality and SOP targets, used for status and reward rules.
`reworkT`	0.04	ratio	Rework ceiling. RCA and status.
`w`	out 0.30 aht 0.20 util 0.20 qual 0.15 sop 0.15	weights	Performance index C01. Must sum to 1.
`uIdleSec, sIdleSec`	120, 60	seconds	Idle thresholds in section 4.
`minGroup`	5	people	Minimum group size for any aggregate shown above associate level.
`workDays`	250	days	Annualisation for V04 and recommendations.
`Capacity-gap targets`	nva 0.08 wait 0.11 uidle 0.04 sidle 0.04	ratio of T03	Attribution of the capacity gap (section 11.2).

6 KPI catalogue

The table lists every KPI shown in the product. IDs are stable. The roll-up column says how the value is computed above associate level; where it says recompute, take the sums of the numerator and denominator and apply the formula again.

Time & session

ID	KPI	Unit	Formula	Roll-up
`T01`	**Logged-in time**	min	Σ (logout_ts − login_ts) over sessions in period	Sum
`T02`	**Break time**	min	Σ duration of state = BREAK	Sum
`T03`	**Available time**	min	T01 − T02	Sum
`T04`	**Active time**	min	VA + BVA + NVA minutes	Sum
`T05`	**Productive time**	min	VA + BVA minutes	Sum
`T06`	**Wait / hold time**	min	Minutes in external wait with no concurrent touch work	Sum
`T07`	**User idle time**	min	Input inactivity ≥ 120 s not explained by call, break or system wait	Sum
`T08`	**System idle time**	min	App unresponsive, no work allocated or portal timeout ≥ 60 s	Sum
`T09`	**Total idle time**	min	T07 + T08	Sum
`T10`	**NVA time**	min	Σ segments classified NVA by the process pack	Sum

Utilisation

ID	KPI	Unit	Formula	Roll-up
`R01`	**Productive time %**	%	T05 ÷ T03	Recompute from sums
`R02`	**Utilisation %**	%	T04 ÷ T03	Recompute from sums
`R03`	**Occupancy %**	%	(T04 + T06) ÷ T03	Recompute from sums

ID	KPI	Unit	Formula	Roll-up
R04	**User idle %**	%	T07 ÷ T03	Recompute from sums
R05	**System idle %**	%	T08 ÷ T03	Recompute from sums
R06	**Total idle %**	%	T09 ÷ T03	Recompute from sums
R07	**NVA %**	%	T10 ÷ T03	Recompute from sums

Output

ID	KPI	Unit	Formula	Roll-up
O01	**Transactions completed**	count	Count of claim_closed events attributed to the associate	Sum
O02	**Average handle time (AHT)**	min	(T04 + T06) ÷ O01	Recompute from sums
O03	**Expected output**	count	T03 × target occupancy ÷ standard AHT	Sum of associate values
O04	**Output attainment %**	%	O01 ÷ O03	Recompute from sums
O05	**AHT attainment %**	%	min(1, standard AHT ÷ O02)	Recompute from sums, cap applied after aggregation
O06	**Transactions per available hour**	count/h	O01 ÷ (T03 ÷ 60)	Recompute from sums
O07	**Productivity gap**	count	max(0, O03 − O01)	Sum of associate shortfalls, not netted
O08	**Capacity gap**	min	O07 × standard AHT	Sum

Quality

ID	KPI	Unit	Formula	Roll-up
Q01	**Quality score**	%	Σ(audit score × claims audited) ÷ Σ claims audited	Weighted by claims audited
Q02	**SOP adherence**	%	Compliant SOP checkpoints ÷ checkpoints evaluated	Weighted by checkpoints evaluated
Q03	**Rework rate**	%	Claims reopened within 72 h ÷ O01	Recompute from sums
Q04	**SLA at risk**	count	Open claims with elapsed ≥ 80% of SLA	Sum

Value

ID	KPI	Unit	Formula	Roll-up
V01	**NVA by cause**	min	T10 split by cause code from the process pack	Sum
V02	**Application switching**	count, min	Foreground application changes per transaction; time lost per change	Recompute from sums
V03	**Excel dependency**	min/day	Minutes in spreadsheet applications per associate-day	Mean per associate-day
V04	**Automation opportunity**	h/year	Cause minutes × associates × working days ÷ 60	Sum
V05	**FTE opportunity**	FTE	Σ excess minutes above target ÷ available minutes per FTE	Recompute from sums
V06	**Before vs after**	%	(after − before) ÷ before for each tracked metric	Recompute

Composite

ID	KPI	Unit	Formula	Roll-up
C01	**Performance index**	points	100 × (0.30·O04 + 0.20·O05 + 0.20·min(1, R02÷0.80) + 0.15·Q01 + 0.15·Q02), terms capped at 1	Associate only
C02	**Team index**	points	Σ(C01 × T03) ÷ Σ T03	Weighted by T03

Rewards

ID	KPI	Unit	Formula	Roll-up
G01	**Points**	points	Σ rule points with caps and gates	Sum
G02	**Level**	level	Highest level whose threshold ≤ cumulative points	Associate only
G03	**Badge progress**	%	Qualifying days or claims ÷ badge requirement	Associate only

Nudges

ID	KPI	Unit	Formula	Roll-up
N01	**Nudges issued**	count	Count of nudges sent in period	Sum
N02	**Acknowledgement rate**	%	Acknowledged ÷ sent	Recompute from sums
N03	**Median time to acknowledge**	min	Median of (ack_ts − sent_ts)	Median of nudges
N04	**Recovery rate**	%	Nudges followed by trigger metric back in range within 15 min ÷ sent	Recompute from sums

Risk

ID	KPI	Unit	Formula	Roll-up
F01	**Occurrence rating**	1–10	Lookup of failure frequency per 100 transactions	Register level
F02	**Risk priority number**	1–1000	Severity × Occurrence × Detection	Register level

RCA

ID	KPI	Unit	Formula	Roll-up
A01	**Driver contribution**	min/txn	Actual minutes per transaction by state − standard minutes per transaction	Recompute from sums

6.1 Definitions that cause the most questions

Term	Definition and reasoning
Occupancy vs utilisation	**Utilisation** counts only time the associate was touching work (active ÷ available). **Occupancy** also counts external wait that the associate could not shorten (active plus wait ÷ available). Occupancy is the measure of how full the shift was; utilisation is how much of it was hands-on.
Average handle time	AHT divides active plus wait time by transactions closed. It is a period-level ratio, not an average of per-claim values, so partial work on unclosed claims does not distort it.

Term	Definition and reasoning
System idle vs user idle	System idle is capacity lost to the environment and is never shown as an associate shortcoming or used to trigger an associate nudge. User idle is a behavioural input gap not explained by calls, breaks or system waits. Both count towards total idle, shown separately.
Wait vs hold	Wait or hold is counted as its own state only when nothing else is being worked. If the associate starts a second claim during a payer hold, those minutes are credited to the second claim. This is why the parallel-work nudge increases output without changing logged time.
Expected output	Available time multiplied by target occupancy and divided by standard handle time. It is the number of transactions an associate would close if the shift ran at target occupancy and standard AHT.
Capacity gap	Expected output less actual output, expressed in minutes by multiplying the shortfall by standard AHT. Shortfalls only; an associate above expectation does not offset another below it.
Output attainment cap	The performance index caps output attainment at 100 percent so that speed beyond expectation cannot offset low quality or SOP adherence. The raw ratio is still reported as O04.

7 Worked example

The associate view in the prototype shows one day for the sample associate, Anita Raman, as of 13:05. The table gives the inputs, each calculation and the result. Use it as the first regression test for the metrics layer.

Inputs

Input	Value	Input	Value
Logged in	303 min	Wait / hold	28 min
Break	30 min	User idle	21 min
Value-add	130 min	System idle	17 min
Business value-add	41 min	Transactions closed	25
Non value-add	36 min	Quality (audited)	96.2%
Reopened claims	1	SOP adherence	94.0%

Check: 30 + 130 + 41 + 36 + 28 + 21 + 17 = 303, equal to logged time.

Calculation

ID	KPI	Calculation	Result
T03	Available time	303 − 30	273 min
T04	Active time	130 + 41 + 36	207 min
T05	Productive time	130 + 41	171 min
T09	Total idle	21 + 17	38 min
R01	Productive time %	171 ÷ 273	62.6%
R02	Utilisation %	207 ÷ 273	75.8%
R03	Occupancy %	(207 + 28) ÷ 273	86.1%
R04	User idle %	21 ÷ 273	7.7%
R05	System idle %	17 ÷ 273	6.2%
R06	Total idle %	38 ÷ 273	13.9%
R07	NVA %	36 ÷ 273	13.2%

ID	KPI	Calculation	Result
O02	Average handle time	(207 + 28) ÷ 25	9.40 min
O03	Expected output	273 × 0.85 ÷ 9.0	25.78
O04	Output attainment	25 ÷ 25.78	97.0%
O05	AHT attainment	min(1, 9.0 ÷ 9.40)	95.7%
O06	Transactions per available hour	25 ÷ (273 ÷ 60)	5.49
O07	Productivity gap	max(0, 25.78 − 25)	0.78
O08	Capacity gap	0.78 × 9.0	7.05 min
Q03	Rework rate	1 ÷ 25	4.0%
C01	Performance index	100 × (0.30×0.970 + 0.20×0.957 + 0.20×0.948 + 0.15×0.962 + 0.15×0.940)	95.7

Utilisation component for the index: min(1, 0.758 ÷ 0.80) = 0.948. Unrounded terms give 95.72.

8 Performance index

The index (C01) gives an associate one number on a 0 to 100 scale. It blends five measures, each capped at 1 so that no single measure can carry the score.

```
out   = min(1, O04)
aht   = min(1, stdAht ÷ O02)
util  = min(1, R02 ÷ utilT)
qual  = Q01 ÷ 100
sop   = Q02 ÷ 100

C01 = 100 × (w.out·out + w.aht·aht + w.util·util + w.qual·qual + w.sop·sop)
```

Component	Weight	Rationale
Output attainment	0.30	Primary measure of throughput against what the shift should deliver.
AHT attainment	0.20	Protects against output gained by working fewer, easier claims.
Utilisation	0.20	Capped at target so that rushing past 80 percent is not rewarded.
Quality	0.15	Audited score.
SOP adherence	0.15	Compliance share. Reward gates in section 9 add a hard floor.

Bands used for status

Audience	Band	Label
Associate	95 or more 85 to under 95 Under 85	On target Near target Below target
Supervisor and leadership, team level	92 or more 88 to under 92 Under 88	Ahead On track Needs support

Team index (C02) is the available-time weighted mean of associate indexes. Leaders see teams only; individual rank is never shown to peers. Associates see a band relative to their team ("top 25 percent"), never a rank or a name.

9 Rewards: points, levels and badges

Rewards recognise efficiency, SOP compliance, non value-add reduction, quality and consistency. They compare an associate with their own role targets, not with colleagues. Points only increase.

9.1 Principles to enforce in code

- **Compliance gate.** Efficiency (E) and NVA reduction (N) points are held for the day if SOP adherence is under 90 percent or quality is under 90 percent. This stops speed being rewarded at the cost of compliance.
- **No deductions.** A missed target earns nothing. The ledger has no negative rows.
- **Daily cap.** The sum of finalised daily points is capped at 300.
- **Provisional then final.** Points are recalculated every five minutes and shown as earned. Consistency and team points, and any day affected by an open dispute, stay pending until day end review.
- **Explainable.** Every ledger row stores the rule, the measured value and the target so the associate can see why points were awarded or not.

9.2 Point rules

ID	Category	Rule	Points	Rule cap
E1	Efficiency	Hourly output at or above 95% of hourly expected	10 per hour	80
E2	Efficiency	Day AHT at or below standard	20	20
E3	Efficiency	Occupancy between 82% and 90%	20	20
S1	SOP	SOP adherence at or above 95%	40 (+20 at 98%)	60
S2	SOP	No critical SOP deviation	20	20
N1	NVA reduction	NVA at or below 10% of available time	30	30
N2	NVA reduction	NVA lower than your own 7-day average	5 per point of reduction	25
Q1	Quality	Audited quality at or above 95%	30	30
Q2	Quality	No reopened claim in last 20	20	20
C1	Consistency	Streak of qualifying days (credited at day end)	10 per day, max 5 days	50
T1	Team	Contribution to team challenge	up to 20	20

Rule details

```
E1  hourExp = O03 ÷ (T03 ÷ 60)    expected completions per available hour
    hour qualifies if completed in the hour ≥ 0.95 × hourExp
    only completed clock hours count
E3  occupancy R03 between 0.82 and 0.90 inclusive
S1  Q02 ≥ 95 → 40 points; Q02 ≥ 98 → 60 points (not additive)
N1  R07 ≤ 0.10
N2  min(25, 5 × floor(avg7dNVA% − todayNVA%)), when positive
C1  10 per consecutive qualifying day, max 5 days, at day end
    qualifying day: C01 ≥ 90 and Q02 ≥ 90
```

Example: the sample day

Rule	Evidence	Points	Status
E1	Hourly expectation 5.67, threshold 5.38. Completed hours: 3, 5, 6, 5, 5. Only the 6 qualifies: 1 of 5 hours.	10	Earned
E2	AHT 9.40 min is above the 9.0 standard.	0	Not earned
E3	Occupancy 86.1% is within 82 to 90%.	20	Earned
S1	SOP adherence 94.0% is under 95%.	0	Not earned
S2	No critical SOP deviation recorded.	20	Earned
N1	NVA 13.2% is above 10%.	0	Not earned
N2	NVA 13.2% against a 7-day average of 14.8%. floor(1.6) = 1, 5 × 1.	5	Earned
Q1	Audited quality 96.2% is at or above 95%.	30	Earned
C1	Six-day streak, 50 points credited at day end.	50	Pending
T1	Team challenge contribution.	12	Pending
	Earned today	**85**	+62 pending

9.3 Levels

Level is the highest level whose threshold is at or below cumulative points. Cumulative points never reset. The sample associate has 2,840 points, which is level 4 (Specialist), 360 points from level 5.

Level	Name	Cumulative points from
1	Associate	0
2	Practitioner	400
3	Proficient	1000
4	Specialist	1800
5	Senior Specialist	3200
6	Expert	5000
7	Master	8000

9.4 Badges

Badges are evaluated by an event-driven job after day close and, for claim-count badges, on each closing event. Each badge is awarded once per associate. Bonus points are written to the ledger at award time. Progress (G03) equals qualifying days or claims divided by the requirement.

Badge	Tier	Category	Requirement	Bonus	Window
Pace Setter	Gold	Efficiency	AHT at or below standard for 5 consecutive days	120	5 days
Throughput Pro	Silver	Efficiency	Output attainment of 100% or more across a full week	100	n/a
Flow Keeper	Silver	Efficiency	Occupancy between 82% and 90% for 5 shifts	80	n/a
Precision Pro	Gold	SOP	SOP adherence of 98% or more for 5 days	150	n/a
Zero Deviation	Platinum	SOP	No SOP deviation across 100 claims	250	100 claims

Badge	Tier	Category	Requirement	Bonus	Window
Audit Ready	Silver	SOP	Complete documentation on 50 consecutive audited claims	100	50 claims
Lean Operator	Gold	NVA reduction	NVA at or below 10% of available time for a full week	150	7 days
Tracker Free	Bronze	NVA reduction	A full day with under 5 minutes of manual tracker work	60	1 days
Clean Close	Silver	Quality	No reopened claim in 20 consecutive closes	100	n/a
First-Time Right	Gold	Quality	Audited quality of 98% or more for the month	200	22 days
Steady Hand	Silver	Consistency	Performance index of 90 or more for 10 days	100	10 days
Always On	Bronze	Consistency	100% schedule adherence for 20 days	80	n/a

Consecutive-day and consecutive-claim requirements reset on the first miss. Disputed minutes are excluded until review, so a pending dispute pauses, not breaks, a streak.

10 Nudge rules

Nudges are rules fired on live signals and shown to the supervisor with the evidence. A nudge is sent to an associate only when the supervisor approves it. Condition-level and system-caused signals go to the supervisor as team items.

Rule	Trigger (default)	Severity	KPI link	Suggested action
User idle	No input for 5 minutes or more, no call active, no system condition	Medium	T07	Check in
Long hold	On payer hold for 10 minutes or more with no second claim open. High at 15 minutes.	Medium or High	T06	Suggest parallel work
SOP step skipped	A checkpoint in the process pack failed before close	Medium	Q02	Send reminder
AHT drift	Last three claims average 20 percent or more above standard AHT	Medium	O02	Offer a review
Spreadsheet streak	Spreadsheet application in the foreground for 8 of the last 30 minutes	Low	V03	Share tip
SLA ageing	Open claim has used 80 percent of its SLA	High	Q04	Prioritise claim
System latency	Three or more associates wait 60 seconds or more on the client application within 10 minutes	Team	T08	Raise IT incident
Recognition	Five consecutive days at or above 100 percent output attainment	Positive	O04	Send recognition

Governance rules

- One nudge per associate per rule per 60 minutes (default cooldown). A nudge is suppressed for 10 minutes after a break and in the first 15 minutes of a shift.
- Associates never receive a nudge for system idle or for wait that is external to them.
- The supervisor can edit the message before sending. The edited text, sender and time are stored with the nudge.
- An associate acknowledges with one action, or the system records an acknowledgement when the corrective event occurs.

Nudge effectiveness KPIs

```
N01  sent today = count(nudges with status in {sent, acknowledged})
N02  acknowledged within 5 min ÷ sent
N03  median(ack_ts − sent_ts)
N04  nudges where the trigger metric returned to range within 15 min ÷ sent
```

11 Root-cause analysis formulas

The supervisor workbench decomposes a problem into measured drivers. Each decomposition is additive so that the drivers add back to the headline number.

11.1 AHT above standard

```
m(s)   = minutes in state s ÷ O01      s in {va, bva, nva, wait}
var(s) = m(s) − stdMix(s)
AHT − stdAht = Σ var(s)       (exact, because Σ stdMix = stdAht)
cause min per txn = cause minutes ÷ O01   (NVA and wait causes)
```

Idle states are not part of AHT, so they do not appear here. They appear in the capacity-gap decomposition below.

11.2 Capacity gap

For each associate with a shortfall, the gap is distributed across the states that exceeded their target share of available time, in proportion to the excess.

```
gap(a)      = O08(a)                                  minutes
excess(a,s) = max(0, minutes(a,s) − target(s) × T03(a))
              for s in {nva, wait, uidle, sidle}
share(a,s)  = gap(a) × excess(a,s) ÷ Σ excess(a,·)   (0 if Σ = 0)
driver(s)   = Σ share(a,s) ÷ associates
```

> **Interpretation.** The attribution is proportional and indicative, not causal proof. The user interface labels it as "attributed to the states that exceeded their target share". Targets are the capacity-gap targets in section 5.

11.3 Rework above ceiling

Rework drivers use the reopen reason code captured when a claim is reopened. Reason share = claims reopened for the reason ÷ claims reopened. The sample reasons are code mismatch, missing call reference, payer reopened, wrong claim attributed and other. Reason codes belong to the process pack.

11.4 Expected effect of an action

Unless a recommendation states otherwise, the expected effect assumes 50 percent capture of the measured excess:

```
effect = capture × max(0, driver − standard)
default capture 0.5, automation 0.8
```

12 FMEA scoring

The failure mode and effects analysis ranks process risks by severity, occurrence and detection. Occurrence is derived from observed frequency, so the register updates itself as the data changes.

```
RPN        = Severity × Occurrence × Detection     (1 to 1,000)
```

```
Occurrence = lookup(frequency per 100 transactions)
frequency  = failure-mode events ÷ 001 × 100
```

Frequency per 100 transactions	Occurrence	Frequency per 100 transactions	Occurrence
under 0.1	1	4 to under 7	6
0.1 to under 0.5	2	7 to under 10	7
0.5 to under 1	3	10 to under 15	8
1 to under 2	4	15 to under 25	9
2 to under 4	5	25 or more	10

Severity and detection scales

Score	Severity (impact if it happens)	Detection (how likely it is caught)
9 to 10	Regulatory or privacy breach, or timely-filing loss	Not detectable, or no automated control
7 to 8	Claim reopened, cash delayed, audit failure	Detected only in periodic audit
4 to 6	Rework or lost capacity for the team	Detected within the day by a report
1 to 3	Minor effect with easy recovery	Detected in real time by a rule

Action priority

```
High    if Severity ≥ 9  or  RPN ≥ 200  or  (Severity ≥ 7 and Occurrence ≥ 7)
Medium  else if RPN ≥ 100
Low     otherwise
```

A high severity cannot be diluted by good detection, which is why severity of 9 or more is High regardless of RPN. Severity and detection are set by the process owner; occurrence is computed but can be overridden in the simulator, in which case the override and user are stored.

Risk matrix and loss pareto

- Matrix cell: row = ceil(Severity ÷ 2), column = ceil(Occurrence ÷ 2). Shading is proportional to row band × column band. Failure-mode IDs are listed in the cell.
- Loss pareto: minutes lost per associate-day by cause = cause minutes ÷ associates in scope ÷ days in period. Sorted descending with a cumulative percentage. Each cause has a type (process, system, behaviour, external) and a controllable flag.
- Controllable means within the organisation or its client IT to change. Payer hold and payer portal wait are external. Client application wait is controllable by the client IT owner.
- Cause by team: the same minutes per associate-day, per team, with heat shading normalised within each cause column.

13 Recommendations and FTE opportunity

Recommendations are made for teams and supervisors, never individuals. Each has a measured size, an effort rating and a confidence, and is ranked by priority score.

```
minutes saved per associate-day = cause min per associate-day × capture
FTE            = minutes saved × associates ÷ mean available min
hours per year = minutes saved ÷ 60 × associates × workDays
priority score = FTE × confidence ÷ effort
```

```
                effort 1 (low) to 5 (high), confidence 0 to 1
```

FTE opportunity (V05)

```
excess(a) = max(0, nva(a) − nvaT × T03(a))
          + max(0, uidle(a) + sidle(a) − idleT × T03(a))
V05       = Σ excess(a) ÷ mean T03
Per 100 seats = V05 ÷ associates × 100
```

Not netted. Opportunities overlap, for example tracker automation and application switching. The sum shown is gross and is labelled "not netted for overlap". A netted figure requires a joint model and is out of scope for v1.

Sample recommendation set

ID	Recommendation	Capture of cause	Effort	Confidence	Link
R1	Automate tracker write-back from claim events	0.70 of tracker updates	3	0.80	F01
R2	Parallel-work protocol and call-block scheduling for payer holds	0.30 of payer hold	2	0.70	F10
R3	Integrate payer portal with client application	0.75 of duplicate entry	4	0.65	F06
R4	Client application performance fix and service target	0.60 of application latency	4	0.50	F05
R5	Single-pane view for application, portal and tracker	0.50 of switching	5	0.50	Switching
R6	Floor-walk cadence for occupancy dip	0.20 of user idle	1	0.60	User idle
R7	Make call reference and note template required at close	0.50 of rework minutes	2	0.75	F02, F07
R8	Targeted SOP coaching for teams below 94 percent	0.05 of user idle plus 0.20 of rework minutes	1	0.70	F03

Action register

Actions come from RCA, coaching and recommendations. Fields: action_id, title, source, owner, due date, status (Proposed, Approved, In progress, Delivered, Rejected), escalated flag and team. A supervisor can escalate a team action to leadership. Delivered actions are verified against the before measure (`V06`): the tracked metric is captured at approval and compared at 4 weeks after delivery.

14 Aggregation, privacy and access

14.1 Aggregation rules

- Sum minutes and counts. Recompute every ratio from the sums. Never average percentages.
- Index is weighted by available time. Quality and SOP are weighted by claims audited and checkpoints evaluated.
- Productivity gap and FTE excess are computed per associate then summed. A surplus never offsets a shortfall.
- For week and month, per-associate averages divide by associate-days, not by calendar days.

14.2 Access model

Role	Can see	Cannot see
Associate	Own metrics, own timeline, own idle episodes, own points and badges, team median and a band relative to the team	Any other associate by name or rank
Supervisor	Associates in direct teams, including individual rows, nudges, coaching sheets	Associates outside the team. Other supervisors' individual data.
Leadership	Teams and supervisors, FMEA, recommendations, actions	Individual associate rows. Teams under five people are suppressed.

14.3 Privacy controls

- No screen content, keystrokes or clipboard are captured. Only window class, allow-listed title patterns and input activity.
- No protected health information is stored. Claim and patient identifiers are replaced by vault tokens on the device.
- Minimum group size of five for any aggregate above associate level. Where a filter would produce a smaller group, return a suppression flag, not a number.
- Every access to an individual record is logged with user, purpose and time. Retention follows the client contract: raw events 90 days, daily facts 25 months.
- Works council or local labour-law consultation is a deployment prerequisite in each jurisdiction. Out of scope here.

15 API contract

Endpoints are versioned under `/v1`. All responses return values and the sums behind ratios. Dates use ISO 8601, times use UTC with an offset. Paging is cursor based above 200 rows. Live endpoints use server-sent events with a 2 second reconnect back-off.

Persona	Widget	KPI IDs	Endpoint	Refresh
Agent	Performance index ring and components	C01, O04, O05, R02, Q01, Q02	`GET /v1/agents/me/index`	60 s
Agent	KPI tiles (8)	O01, O02, R03, R02, R01, R07, R04, R05	`GET /v1/agents/me/summary?period=`	60 s
Agent	Session strip	T01, T02, T03, T04, T06, T09, Q03	`GET /v1/agents/me/session`	60 s
Agent	Time distribution donut	T02, T05, T06, T07, T08, T10	`GET /v1/agents/me/time-distribution`	60 s
Agent	Shift timeline	State segments	`GET /v1/agents/me/timeline`	30 s
Agent	Output per hour	O01, O03	`GET /v1/agents/me/hourly`	5 min
Agent	Idle analysis and episodes	T07, T08, R04, R05	`GET /v1/agents/me/idle · POST /v1/agents/me/idle/{id}/dispute`	60 s
Agent	Benchmarks	R01–R07, O02	`GET /v1/agents/me/benchmarks`	Daily
Agent	SOP checkpoints	Q02	`GET /v1/agents/me/sop`	Hourly
Agent	Points, level, streak, badges	G01, G02, G03	`GET /v1/agents/me/rewards/*`	5 min

Persona	Widget	KPI IDs	Endpoint	Refresh
Supervisor	Team KPI tiles (10)	C02, O01, O02, R03, R02, R01, R07, R06, Q01, Q04	`GET /v1/teams/{id}/summary?period=`	60 s
Supervisor	Live floor	Status per associate	`GET /v1/teams/{id}/floor (SSE)`	Push, 15 s
Supervisor	Throughput and occupancy heatmap	O01, O03, R03	`GET /v1/teams/{id}/hourly · /occupancy-heatmap`	5 min
Supervisor	Associate table	C01 and 11 KPIs per associate	`GET /v1/teams/{id}/associates`	60 s
Supervisor	NVA pareto, wait and system causes	V01, T06, T08	`GET /v1/teams/{id}/nva-causes · /wait-causes`	5 min
Supervisor	Nudge feed and statistics	N01, N02, N03, N04	`GET /v1/teams/{id}/nudges (SSE) · POST /nudges/{id}/send`	Push
Supervisor	RCA workbench	A01, O08, Q03	`GET /v1/teams/{id}/rca?problem=`	On demand
Supervisor	Coaching queue and 1:1 sheet	C01 and flags	`GET /v1/teams/{id}/coaching-queue`	Hourly
Leadership	Organisation KPI tiles (10)	C02, O04, O02, R03, R02, R07, R06, Q01, O08, V05	`GET /v1/org/summary?period=`	5 min
Leadership	Capacity gap by driver	O03, O07, O08, V05	`GET /v1/org/capacity-gap`	5 min
Leadership	Supervisor scorecard and comparison	C02 and 10 KPIs per team	`GET /v1/org/teams/scorecard`	5 min
Leadership	FMEA table, simulator, matrix	F01, F02	`GET /v1/risk/fmea · PUT /v1/risk/fmea/{id}/scores`	On demand
Leadership	Loss pareto and cause by team	V01, T06, T07, T08	`GET /v1/org/loss-pareto · /loss-by-team`	Daily
Leadership	Recommendations and impact matrix	V04, V05, V06	`GET /v1/recommendations`	Daily
Leadership	Action register	Action status	`GET /v1/actions · PATCH /v1/actions/{id}`	On demand

Example response

`GET /v1/agents/me/summary?period=today` for the sample day:

```
{
  "associate": "AS-4417", "as_of": "13:05", "period": "today",
  "time": { "logged_min": 303, "break_min": 30, "available_min": 273,
            "active_min": 207, "productive_min": 171, "wait_min": 28,
            "user_idle_min": 21, "system_idle_min": 17, "nva_min": 36 },
  "output": { "completed": 25, "expected": 25.8, "aht_min": 9.4,
              "output_attainment": 0.970 },
  "ratios": { "occupancy": 0.861, "utilisation": 0.758, "productive": 0.626,
              "nva": 0.132, "user_idle": 0.077, "system_idle": 0.062 },
  "quality": { "quality": 96.2, "sop": 94.0, "rework_rate": 0.040 },
  "index": 95.7
}
```

Errors and states

- `200` with `"suppressed": true` where a group is below the minimum size.

- `200` with `"status": "partial"` and `as_of` when a source is late. The UI shows the as-of time and never an empty tile.
- `403` for a scope violation. `404` is not used to hide existence of a team.

16 Validation test cases

Each case has an input, an expected output and the KPI it covers. The first is the worked example in section 7.

#	Scenario	Expected	Covers
1	Sample day inputs from section 7	Every value in the section 7 calculation table, tolerance ±0.01 on ratios	T, R, O, C01
2	Time-state reconciliation for any associate-day	logged = brk + va + bva + nva + wait + uidle + sidle	Invariant
3	Two claims open, payer hold on A for 12 min while B is worked	Hold minutes credited to B as touch time. `wait` unchanged.	`T06`
4	Input gap of 119 s, then 121 s, no call	119 s: no idle. 121 s: 121 s counted as user idle from the last input.	`T07`
5	Application unresponsive for 59 s, then 61 s	59 s: not system idle. 61 s: system idle from the start of the condition.	`T08`
6	Input gap during an active call	Not user idle.	`T07`, `call_state`
7	Team of 25: sum of associate indexes weighted by T03	C02 equals weighted mean to 0.01	`C02`
8	Team ratio check: team R03 from sums vs mean of associate R03	Team value equals recompute from sums, not the mean	Roll-up
9	Output above expected (O04 = 1.12)	Index output term is 1.0. O04 reports 112%.	`C01`
10	SOP adherence 89% on the day	E and N rules held. Q, S2 still evaluated.	Gate
11	AHT 9.0 exactly	E2 awarded (at or below standard).	`E2`
12	NVA 14.8% average, today 13.19%	N2 = 5 × floor(1.61) = 5	`N2`
13	Dispute on a 4-minute idle episode, then accepted as wait	Idle −4, wait +4 after review. Points recalculated.	Review
14	Frequency 18 per 100, severity 6, detection 7	O = 9, RPN = 378, priority High	`F01`, `F02`
15	Severity 10, frequency 0.4, detection 7	O = 2, RPN = 140, priority High (severity rule)	Action priority
16	Team with 4 associates	Suppression flag, no value	Privacy
17	Capacity gap, associate A excess nva 20, wait 10, gap 6 min	nva 4.0, wait 2.0 attributed	11.2
18	Recommendation with FTE 7.1, confidence 0.8, effort 3	Score 1.89 (7.1 × 0.8 ÷ 3)	13

17 Assumptions and open items

Assumptions made in this version

- Thresholds, weights, targets and benchmark ranges are placeholders that reflect typical revenue-cycle operations. Industry benchmark ranges shown in the associate view are not from a published survey. Replace them with the client contract or an agreed source.

- Telephony and workflow adapters can emit the events in section 3. Where a client cannot provide `claim_closed`, the product falls back to title-pattern detection and reports lower confidence.
- Quality scores arrive by audit sampling. Q01 is therefore a weighted mean of audited claims, and a day with no audited claims shows the trailing value with a flag.
- The supervisor third tab is "Coaching and actions". The original brief was cut off at that point.

Open items for engineering and design

#	Question	Decision needed from
1	Does the client allow a title-pattern allow-list for all applications in scope, including the payer portals?	Client IT and compliance
2	Reopen window for Q03: 72 hours as modelled, or per contract?	Process owner
3	Should overlap between recommendations be netted using a joint model in v2?	Operations excellence
4	Retention periods and works-council requirements by location	Legal and HR
5	Should the daily points cap and the compliance gate be configurable per client?	Product owner

Appendix A Seed failure-mode register

Failure modes shown in the FMEA view with sample frequencies. Occurrence is computed from frequency using the section 12 table, so changing a frequency changes the score.

ID	Step and failure mode	Effect and control	S	O	D	RPN	Priority	Detection
F01	**Claim status update** Tracker or Excel update omitted	Status out of date, duplicate work later Control: End-of-day reconciliation report	5	8	6	240	High	KPI-V03
F02	**Payer call** Call reference number not recorded	Cannot evidence a payer commitment Control: Real-time SOP rule and nudge	7	7	4	196	High	KPI-Q02
F03	**Eligibility check** Eligibility not verified before the call	Wrong payer contacted, wasted call Control: Weekly SOP audit sample	6	6	5	180	Medium	KPI-Q02
F04	**Claim attribution** Minutes attributed to the wrong claim	Handle time and KPIs distorted Control: Confidence score and Fix my day review	4	5	3	60	Low	KPI-O02
F05	**Application access** Client application latency over 60 s	Lost capacity and SLA risk Control: System-idle alert to IT	6	8	2	96	Low	KPI-T08
F06	**Payer portal** Portal timeout during submission	Re-keying and duplicate entry Control: Timeout detector	5	6	4	120	Medium	KPI-T08
F07	**Documentation** Notes not written to template	Audit failure, poor hand-off Control: Monthly audit sample	6	9	7	378	High	KPI-Q02
F08	**Claim close** Status or denial code mismatch	Claim reopened, cash delayed Control: Reopen report	8	6	6	288	High	KPI-Q03
F09	**Data handling** PHI typed into free-text note	Privacy incident Control: Annual training, manual review	10	2	7	140	High	None automated
F10	**Hold management** Hold over 15 min with no parallel work	Idle capacity during hold Control: Real-time hold nudge	4	8	3	96	Low	KPI-T06
F11	**Queue ageing** Claim exceeds its SLA	Timely-filing risk Control: SLA-at-risk alert	9	4	4	144	High	KPI-Q04